

FedTypo: Prototype-Guided Federated Anti-Money-Laundering Detection Under Heterogeneity, Concept Drift, and Delayed Labels

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Abstract—Cross-institutional anti-money-laundering (AML) detection must learn from directed transaction graphs with rare, evolving patterns and delayed confirmation. FedTypo is a federated temporal graph method in which compact prototypes of confirmed-positive behavior guide client grouping and a de-duplicated registry can adjust local ranking. The backbone is shared; classification heads and online feature statistics remain local. We evaluate IBM AMLworld HI-Small and SAML-D prequentially across ten seeds, with source-account-consistent, label-independent account-hash silos, stochastic confirmation delay, asymmetric alert noise, and controlled onsets. The primary metric is stream-level client-macro area under the precision-recall curve (AUPRC): predictions are concatenated over all evaluated windows within each client and then averaged across clients, so negative-only windows remain represented. Precision at 50 alerts (P@50) is secondary. Across ten paired seeds, FedTypo improves primary AUPRC on IBM in both the natural and controlled-drift streams. On SAML-D it improves over FedAvg, FedProx, CDA-FedAvg, and its registry ablation under drift, but not over Local or FedProto; no comparison is significant in the natural stream. The registry itself yields no reliable early-transfer gain, and none of 24 component-ablation comparisons survives Holm correction. These results support a ranking effect without isolating the proposed mechanism. Raw records remain local, but the method does not claim differential privacy or robustness to malicious clients.

Index Terms—Anti-money laundering, concept drift, delayed-label learning, federated graph learning, financial forensics.

I. INTRODUCTION

ANTI-MONEY-LAUNDERING monitoring is naturally relational: a transaction that appears ordinary in isolation can participate in a fan-in, fan-out, cycle, or scatter-gather pattern. Recent AML benchmarks therefore model transactions as graphs and increasingly treat laundering as a subgraph problem [1], [2]. Collaboration is also attractive because a laundering chain can cross institutional boundaries, but centralizing customer and transaction records is often operationally or legally infeasible.

Federated learning (FL) keeps raw records at their originating institution, yet vanilla averaging is poorly matched to this problem. First, each institution sees a different mixture of suspicious behaviors; this can be more structured than generic feature skew. Second, those mixtures change over

time. Third, reliable labels are delayed by investigation, while immediate alerts are imperfect and the positive class is about one transaction in a thousand. The resulting problem combines cross-silo heterogeneity, graph streams, extreme imbalance, and delayed feedback.

FedTypo tests whether compact representations of confirmed suspicious behavior can connect the heterogeneity and temporal-adaptation problems. Each client summarizes confirmed-positive transaction embeddings by at most four behavioral prototypes. Prototype-set similarity guides aggregation groups. Separately, delayed prototype-relative novelty can admit a candidate to a global registry, whose entries adjust local risk scores before the same behavior is extensively confirmed elsewhere. Controlled staggered onsets permit a direct post-onset evaluation of this early-transfer path.

The contributions are:

- A prototype-guided federated temporal graph model that combines prototype-set client grouping, stability-weighted aggregation, personalized heads, and a de-duplicated behavioral registry.
- A delayed-feedback protocol with two distinct temporal signals: immediate embedding maximum mean discrepancy (MMD) controls aggregation stability, whereas novelty among newly confirmed positives controls registry admission.
- A prequential protocol with online-only normalization, matched initialization, source-account-consistent silos, supported-window analysis, time-consistent drift injection, a common per-round optimizer policy, and inference across ten independent seeds.
- A two-benchmark evaluation with component ablations, primary label-independent silos, an outcome-conditioned partition sensitivity, prototype-fidelity and temporal-support diagnostics, registry-boost and budget sensitivity, and an explicit privacy boundary.

II. RELATED WORK

A. Federated Learning Under Heterogeneity

FedAvg [3] weights local models by sample count but can degrade under non-IID data. FedProx adds a proximal penalty to restrict local deviation [4]; FedBN and related personalization methods retain distribution-specific components locally [5]. Clustered FL learns multiple models for latent client groups rather than forcing one global compromise [6]. FedTypo follows this multi-model view but forms groups from

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sets of suspicious-behavior prototypes, not parameter updates, and periodically couples each group model to a common mean. FedProto communicates class prototypes to reduce heterogeneity without averaging full models [7]; in contrast, FedTypo's positive-only prototypes identify client groups and delayed behavioral novelty, while a shared backbone is still aggregated.

B. Federated Streams, Drift, and Sparse Labels

CDA-FedAvg detects changes in prediction confidence and retrain from a rehearsal memory [8]. SFLEDS is especially relevant: it uses prototypes and cross-client consistency for partially labeled evolving streams [9]. FedPLC reorganizes label-wise communities as client concepts evolve [10]. These works combine prototypes, grouping, and temporal adaptation. FedTypo targets a binary graph-ranking regime in which positives are extremely sparse, reliable labels arrive after a stochastic delay, and shared prototypes represent confirmed-positive behavior rather than a well-populated class. In centralized fraud detection, dual supervision streams have long been used to cope with delayed labels and drift [11]; our protocol adapts this idea to cross-silo graph learning.

C. Graph Learning and Federated AML

Graph neural networks have become central to financial-forensics benchmarks [12], [13], [2]. Early federated AML work established the feasibility of collaborative graph learning without moving raw transactions [14]; recent systems study explicit cross-institution boundary representations [15]. However, keeping raw data local is not a formal privacy guarantee. Model parameters and embeddings may leak membership, attributes, or graph structure [16]. We therefore treat prototypes as compact shared representations, not as proof of privacy, and state the associated threat boundary in Section III-B.

III. PROBLEM AND THREAT MODEL

A. Streaming Objective

Institution $k \in \{1, \dots, K\}$ receives a temporally ordered sequence of directed transaction graphs $G_k^w = (V_k^w, E_k^w)$, one per window w . Transaction $e = (u, v, x_e, t_e)$ has features x_e , latent binary label y_e , and, for positives, typology τ_e . Its reliable label becomes visible at $w + \Delta_e$. A noisy alert a_e is available immediately. At the start of window w , before learning from that window, the client must rank its edges for a fixed investigation budget B .

Let $f_k^w(e)$ denote the score. The primary goal is high stream-level average precision within each client, macro-averaged across clients. Per-window precision and recall among the top B edges are secondary investigation-budget measures, while raw transactions, account identifiers, the local head, and local normalization statistics remain within their source silo. Both $P_k(x, y, \tau)$ and its temporal evolution may differ across clients.

B. Threat Boundary

We assume an authenticated cross-silo deployment with an honest-but-curious coordinator that executes the algorithm but may inspect received backbone parameters, prototypes, stability weights, and registry events. Clients are assumed protocol-compliant; Byzantine updates, prototype poisoning, collusion, availability attacks, and adaptive evasion are outside the evaluated model. Injected "drift" is a controlled distribution shift, not an adversarial-security experiment.

FedTypo provides *data locality*: no raw edge, account identifier, label, personalized head, or running statistic is transmitted. It does not provide differential privacy, secure aggregation, or cryptographic confidentiality for model updates and prototypes. Prototype centroids can still encode sensitive population properties, and the backbone remains subject to the usual FL leakage channels [16]. Deployment in a stronger threat model would require secure aggregation for compatible updates plus a separately analyzed protection mechanism for prototypes and registry events. Formal privacy accounting for repeated, data-dependent prototype releases remains future work.

IV. FEDTYPO

A. Temporal Edge Encoder and Delayed Supervision

Each account has a zero-initialized state $m_v^0 \in \mathbb{R}^d$, $d = 64$; there are no static node features or learned account identifiers. The 11 edge features are online-standardized log amount, a cross-border indicator, sine and cosine time-of-day, and seven payment-channel indicators. The first natural window initializes the local amount statistics and temporal memory and is not scored. Later statistics contain only previously observed windows and are updated after scoring. An edge projection $q_{uv} = \text{ReLU}(W_e x_{uv} + b_e) \in \mathbb{R}^d$ enters two single-head, edge-conditioned attention layers [17]. At layer ℓ ,

$$\begin{aligned} \alpha_{uv}^\ell &= \text{softmax}_{(j,v) \in E_k^w}^\top [h_u^{\ell-1} \| h_v^{\ell-1} \| q_{uv}], \\ g_v^\ell &= \sum_{(u,v) \in E_k^w} \alpha_{uv}^\ell (h_u^{\ell-1} + q_{uv}), \\ h_v^\ell &= \text{ReLU}(W_\ell [h_v^{\ell-1} \| g_v^\ell] + b_\ell), \end{aligned} \quad (1)$$

where $h^0 = m^{w-1}$. Messages follow the directed source-to-destination edges. All transactions in a window are processed simultaneously; multiple incoming edges are attention-pooled before one gated recurrent unit (GRU) cell update per incident account. Inactive account states are unchanged. There is one attention head and no dropout. The transaction representation and local score are

$$z_{uv}^w = [m_u^w \| m_v^w \| q_{uv}] \in \mathbb{R}^{3d=192}, \quad r_k(e) = \sigma(h_k(z_{uv}^w)), \quad (2)$$

where h_k is a personalized $192 \rightarrow 64 \rightarrow 1$ ReLU multilayer perceptron (MLP). Linear and GRU parameters use the seeded PyTorch default uniform initializations; the backbone and GRU are federated, while h_k remains local.

After testing window w , the client trains on all labels confirmed by w and the current alert labels:

$$\mathcal{L}_k^w = \mathcal{L}_{\text{focal}}(\mathcal{D}_{k,\text{conf}}^{\leq w}) + \lambda \mathcal{L}_{\text{focal}}(\mathcal{D}_{k,\text{alert}}^w), \quad \lambda = 0.4, \quad (3)$$

with focal parameters $\gamma = 2$, $\alpha = 0.75$ [18]. One local epoch applies Adam (2×10^{-3}) first to the cumulative confirmed batch and then to the current alert batch, whose loss is scaled by $\lambda = 0.4$. Each batch retains all positives followed by at most 20,000 sampled negatives; there is no within-batch shuffle. Adam state is reset after every window for every comparator. FedAvg, FedProx, and CDA-FedAvg use the actual number of processed examples as their aggregation weight. A fixed benchmark-level channel vocabulary is learned from the public warm-up window and then shared; unseen categories map to the all-zero channel encoding. A deployment would predeclare an institution-approved schema rather than pool private warm-up records. Fig. 1 summarizes the complete data flow.

B. Behavioral Prototypes and Fidelity

Let $\mathcal{C}_k^w$ be client k 's confirmed-positive embeddings visible by w . Once eight confirmed positives are available, Euclidean k -means is fitted with four clusters after L2 normalization. The centroids are renormalized, producing $\mathcal{P}_k^w = \{p_{k1}^w, \dots, p_{kq}^w\}$. The client also stores the 95th percentile of the nearest-centroid cosine distance,

$$\tau_k^w = Q_{0.95} \left(1 - \max_{p \in \mathcal{P}_k^w} \langle \bar{z}_e, p \rangle : e \in \mathcal{C}_k^w \right). \quad (4)$$

The centroids summarize modes in the learned positive-transaction embedding; they are not assigned semantic names and do not consume typology labels. As a post hoc diagnostic, we compare prototype assignments with the benchmark's annotated positive typologies using purity, normalized mutual information (NMI), adjusted Rand index (ARI), and the same-type versus different-type cosine-similarity gap. Downstream ranking performance alone is not treated as evidence that a prototype recovers a named typology.

C. Immediate Stability Signal

The immediate path independently samples at most 256 current and 256 previous transaction embeddings. For their pooled matrix Z , let $b = \text{median}_{i,j} \|Z_i - Z_j\|_2^2 + 10^{-9}$ and $k(x, y) = \exp(-\|x - y\|_2^2 / b)$. We use the biased empirical squared maximum mean discrepancy (MMD), including within-sample diagonal terms,

$$\begin{aligned} \widehat{\text{MMD}}_b^2(A, B) &= \frac{1}{|A|^2} \sum_{a, a' \in A} k(a, a') + \frac{1}{|B|^2} \sum_{b, b' \in B} k(b, b') \\ &\quad - \frac{2}{|A||B|} \sum_{a \in A, b \in B} k(a, b), \end{aligned} \quad (5)$$

following [19]. Once five values are available, a robust score compares the current value with the median and median absolute deviation of the preceding history. The aggregation stability is updated as

$$s_k^w = 0.7s_k^{w-1} + 0.3 \exp[-\max(0, z_{\text{MMD}}^w - 3)]. \quad (6)$$

Thus ordinary variation leaves s_k near one, while unusually large embedding shifts temporarily reduce that client's influence. This signal does not create a registry entry.

D. Delayed Novelty and Registry

Novelty is evaluated *before* the current confirmed positives are absorbed into new prototypes. For positives first confirmed at w ,

$$\nu_k^w = \frac{1}{|\mathcal{N}_k^w|} \sum_{e \in \mathcal{N}_k^w} \mathbf{1} \left[1 - \max_{p \in \mathcal{P}_k^{w-1}} \langle \bar{z}_e, p \rangle > \tau_k^{w-1} \right]. \quad (7)$$

The test is gated until $w \geq 3$ and at least 30 positives have been confirmed. If $\nu_k^w > 0.3$ and at least three novel positives exist, their normalized mean becomes a registry candidate. The server rejects it when its cosine similarity to an existing entry exceeds 0.9.

For the full method only, a registry $\mathcal{R}^w$ adjusts the local score:

$$\tilde{r}_k(e) = \text{clip}_{[0,1]} \left(r_k(e) + \beta \max_{p \in \mathcal{R}^w} \max(0, \langle \bar{z}_e, p \rangle) \right). \quad (8)$$

We preselect $\beta = 0.15$ and define the maximum as zero when $\mathcal{R}^w = \emptyset$. A post hoc re-scoring sensitivity changes only β , without retraining or changing registry admission. FedTypo-NoReg executes the identical prototype construction, stability weighting, and grouped aggregation but omits this inference-time term. It is therefore the direct registry ablation.

E. Prototype-Set Grouping and Aggregation

For prototype sets P and Q , the symmetric Chamfer cosine distance is

$$d_{\text{Ch}}(P, Q) = 1 - \frac{1}{2} \left[\frac{1}{|P|} \sum_{p \in P} \max_{q \in Q} p^\top q + \frac{1}{|Q|} \sum_{q \in Q} \max_{p \in P} q^\top p \right]. \quad (9)$$

When at least three clients have prototypes, average-linkage agglomerative clustering forms two groups; otherwise all clients share one group. Assignments refresh every three windows. A client without prototypes is temporarily assigned to group zero.

Let n_k^w be the cumulative confirmed-positive count and $a_k^w = \max(1, n_k^w) \max(0.05, s_k^w)$. For group g and backbone parameters θ_k^w ,

$$\bar{\theta}_g^w = \frac{\sum_{k:g(k)=g} a_k^w \theta_k^w}{\sum_{k:g(k)=g} a_k^w}, \quad \bar{\theta}^w = \frac{\sum_k a_k^w \theta_k^w}{\sum_k a_k^w}, \quad (10)$$

$$\theta_{g,\text{out}}^w = (1 - \rho) \bar{\theta}_g^w + \rho \bar{\theta}^w, \quad \rho = 0.2. \quad (11)$$

The coupling keeps group-specific models in comparable embedding spaces, which is necessary because prototypes are compared across clients. Algorithm 1 summarizes one sequential round.

F. Communication and Complexity

With 11 edge features on either dataset, the shared backbone contains 42,626 float32 parameters (166.5 KiB in each direction per participating model-aggregation round). The 12,417-parameter head remains local. Four 192-dimensional prototypes add at most 3.0 KiB of uplink because each prototype

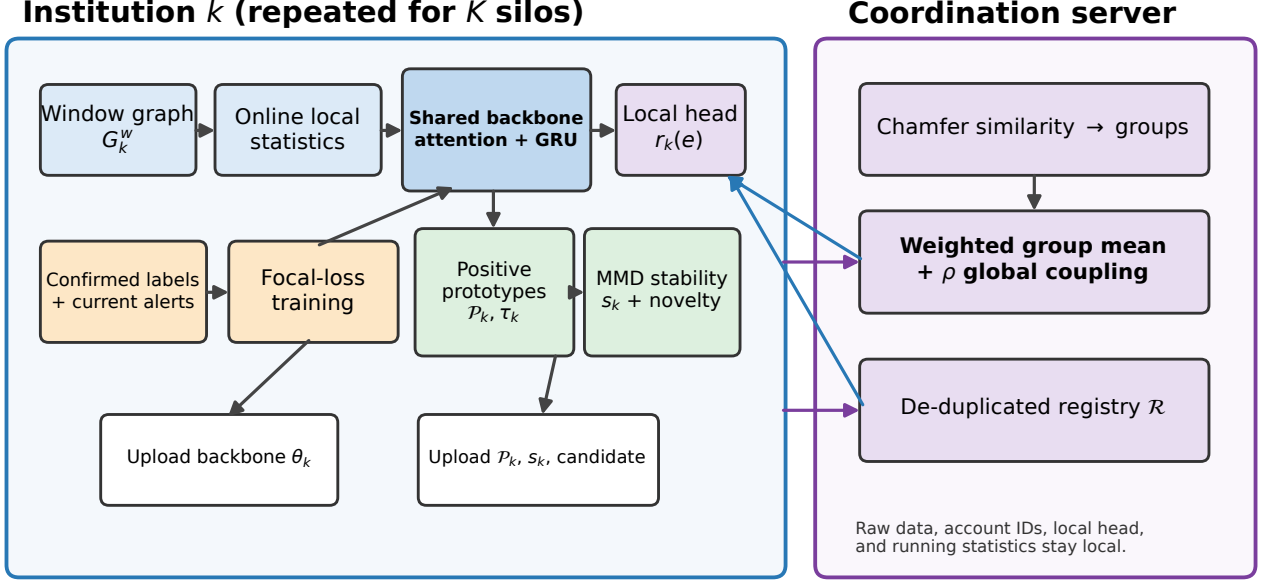


Fig. 1. FedTypo data flow. The immediate MMD path changes aggregation stability; only the delayed novelty path can add a registry entry.

Algorithm 1 FedTypo prequential round w

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- 1: **for** each client k **do**
 - 2: Encode G_k^w , advance incident-account memory, and score
 - 3: Add registry term; update running statistics
 - 4: Compute MMD stability s_k^w
 - 5: Train on delayed confirmations and current alerts
 - 6: Judge newly confirmed positives against $\mathcal{P}_k^{w-1}$
 - 7: Refit $\mathcal{P}_k^w$; send backbone, prototypes, stability, and any registry candidate
 - 8: **end for**
 - 9: De-duplicate registry candidates
 - 10: **if** $w \bmod 3 = 0$ **then**
 - 11: Recluster clients by prototype-set Chamfer distance
 - 12: **end if**
 - 13: Form stability/positive-weighted group means and global coupling
 - 14: Broadcast each group model and the registry
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spans the full $3d = 192$ -dimensional edge embedding, approximately 1.8% of the backbone payload. Registry broadcasts are 0.75 KiB per entry. Local message passing is linear in window edges for fixed d ; exhaustive server matching costs $O(K^2 m^2 d)$, small for $K = 5, m \leq 4$ but a limitation for large federations. The FedProto comparator does not exchange a backbone; it sends up to two 192-dimensional class prototypes (1.5 KiB) plus class supports. Local-only has no communication.

V. EXPERIMENTAL PROTOCOL

A. Datasets and Silo Construction

IBM AMLworld HI-Small contains 5,078,345 transactions, 515,088 accounts, 5,177 positives (0.102%), eight named laundering typologies plus an unstructured positive category over approximately 18 days [1]. SAML-D contains 9,504,852 transactions, 855,460 accounts, 9,873 positives (0.104%), and 17 suspicious types over approximately 320 days [20]. Natural windows are 24 hours for IBM and 17 days for SAML-D. A label-independent support rule removes only a contiguous terminal suffix whose window count is below $\max(1000, \lceil 0.01 \tilde{N} \rceil)$, where $\tilde{N}$ is the median natural-window transaction count. The rule and its thresholds were fixed before the reported runs. All 19 SAML-D windows pass. For IBM, windows 10–17 contain only 1,108 transactions (0.022% of the data) and fail; the ten supported windows retain 5,077,237 transactions, 515,078 accounts, and 4,522 positives. Thus the excluded tail contains 655 positives, or 12.65% of all IBM positives, despite its small transaction share. Window 0 initializes client-local online statistics and temporal memory without contributing an evaluation score, leaving nine IBM and 18 SAML-D evaluation windows. We report the excluded tail and per-window counts rather than assigning an incomplete day the weight of a full window.

The primary partition assigns every source account to one of five clients by a stable seed-keyed hash of its identifier. It requires neither labels, typologies, the complete account universe, nor a graph built from future edges. All outgoing transactions of that source remain together; destination accounts can appear in multiple silos. This construction is a leakage-resistant simulation, not evidence that random hashed silos reproduce bank boundaries.

As a separate non-IID sensitivity, we use a controlled *typology-skew* construction. Louvain communities from the full undirected graph are load-balanced and suspicious source accounts are then reassigned using client-specific Dirichlet(0.3) profiles of their dominant annotated typology [21]. This sensitivity is explicitly transductive and outcome-conditioned; it is not used for the primary inferential claims. Typology labels otherwise enter only post hoc fidelity and intervention diagnostics, never model inputs or pre-confirmation labels.

B. Delayed Labels, Alerts, and Drift

Each transaction receives an independent rounded log-normal confirmation delay with median three evaluation cycles and 95th percentile 12 cycles. Calendar duration therefore differs across datasets, so results are interpreted in normalized review cycles. The immediate alert has true positive rate 0.60 and false positive rate 0.01; this asymmetric process avoids the unrealistic false-alert volume created by symmetric label flips.

The control condition C0 preserves the natural order. In C1, the most prevalent named typology is given an abrupt onset at the dominant client by re-timing its pre-onset examples uniformly to on/after the midpoint. The unstructured positive category is not eligible for drift targeting. The second most prevalent named typology is re-timed at its three highest-mass clients with staggered onsets at 30%, 60%, and 85% of the supported span (rounded to distinct evaluation windows). Transaction count and time-of-day offset are preserved globally, and both timestamp and window are updated. This construction tests adaptation to a controlled change in typology support; it does not synthesize unseen criminal behavior. Each moved transaction and event is tagged for the post-onset mechanism analysis.

C. Comparators, Metrics, and Inference

All methods use the same encoder, local head, loss, online normalization, alerts, confirmation delays, initial backbone, and prequential order:

- Local: independent client training, no communication.
- FedAvg [3]: processed-example-weighted aggregation.
- FedProx [4]: FedAvg with $\mu = 10^{-3}$.
- FedProto [7]: an adaptation that keeps each backbone local, federates confirmed-label class prototypes, and applies a representation penalty of 0.1; noisy alert labels do not construct or regularize prototypes.
- CDA-FedAvg [8]: confidence change detection and rehearsal. Our synchronous, window-level implementation adapts the original asynchronous method.
- FedTypo-NoReg and FedTypo: the direct registry ablation pair.

The window-adapted CDA detector uses a three-window minimum split, sensitivity 0.05, five rehearsal rounds after detection, and a 2,000-negative rehearsal cap while retaining all confirmed positives. No held-out validation split or automated hyperparameter search was used. Exploratory single-seed development runs on both benchmark streams informed

TABLE I
SHARED EXPERIMENTAL CONFIGURATION.

Setting	Value
Clients / seeds	5 / 10 (42–51)
Natural / supported / evaluated windows	IBM 18 / 10 / 9; SAML-D 19 / 19 / 18
Hidden / transaction embedding	64 / 192
Local epochs per window	1
Optimizer / learning rate	Adam / 2×10^{-3}
Focal (α, γ)	(0.75, 2)
Alert weight / TPR / FPR	0.4 / 0.60 / 0.01
Delay median / 95th pct.	3 / 12 cycles
Prototype count / support	4 / ≥ 8 positives
FedProto representation penalty	0.1
Novelty / duplicate gates	0.30 / 0.90 cosine
Group refresh / coupling	3 cycles / $\rho = 0.2$
Secondary alert budget	50 per client-window

the configuration before the ten-seed runs were frozen. In that development stage, the maximum prototype count changed from six to four and the novelty gate from 0.5 to 0.3 to accommodate sparse confirmed positives; the FedProx coefficient changed from 0.01 to 0.001 under the common optimizer protocol. No value was selected separately by dataset or re-tuned after inspecting the reported ten-seed comparisons. The resulting inference measures stochastic repeatability at this frozen configuration, not hyperparameter-selection uncertainty or performance on an untouched external benchmark.

For each seed and method, we concatenate all evaluated-window predictions within a client, compute that client’s area under the precision–recall curve (AUPRC), and average equally across the five clients. This stream-level client macro-average is primary; it neither drops negative-only windows nor lets high-volume clients dominate. As secondary operational summaries, we compute precision among the top 50 alerts in every client-window and macro-average first over windows within a client and then over clients. Budgets 10, 25, 50, 100, and 200 form a sensitivity analysis. Separately, $\beta \in \{0, 0.05, 0.10, 0.15, 0.20, 0.30\}$ re-scores a common FedTypo trajectory on SAML-D; this descriptive analysis is not used to select the predeclared value. Positive-only window AUPRC and transaction-weighted window AUPRC are retained as diagnostics, not primary endpoints. Every condition/method is rerun for seeds 42–51.

Each seed yields one summary per metric. Tables report the mean and 10,000-resample percentile bootstrap 95% confidence interval across those ten seed summaries. FedTypo is paired with each baseline by seed using a two-sided Wilcoxon signed-rank test. Holm adjustment is applied across the six baselines within each dataset-condition family [22]. Client-windows are never treated as independent replicates.

The evaluation addresses three questions. RQ1 asks whether federation and prototype grouping improve client-macro ranking. RQ2 asks whether the registry itself helps under staggered onsets, using FedTypo-NoReg and post-onset target-typology recall. RQ3 asks whether the learned prototypes agree with annotated typologies and whether the ranking transfers across dataset and investigation budget. Table I lists the frozen configuration.

VI. RESULTS

A. Primary Account-Hash Evaluation

Table II reports the primary IBM results.

On IBM, FedTypo obtains AUPRC 0.0036 in C0 and 0.0037 in C1. Every paired comparison with the six baselines has $p_H = 0.012$ after Holm correction and 10/10 seed-level wins. In C1, for example, FedAvg reaches 0.0013 and FedTypo-NoReg 0.0013, corresponding to 191.4% and 189.3% relative AUPRC gains. The secondary budget measure is less favorable: Local has the highest IBM P@50 in both C0 (0.0142 versus 0.0137) and C1 (0.0143 versus 0.0135). The result is therefore a threshold-free ranking gain, not an improvement at the prespecified 50-alert cutoff. Table III gives the corresponding SAML-D results.

On SAML-D C0, no FedTypo comparison is significant after correction and Local has the highest mean AUPRC and P@50. Under C1, FedTypo reaches 0.0020 AUPRC. It is higher than FedAvg (51.2%, $p_H = 0.020$), FedProx ($p_H = 0.012$), CDA-FedAvg ($p_H = 0.020$), and FedTypo-NoReg ($p_H = 0.023$), but not Local or FedProto (both $p_H = 0.387$). Local again has the higher P@50 (0.0316 versus 0.0162). Thus the SAML-D evidence is condition- and comparator-dependent rather than a uniform cross-dataset advantage. Table IV reports the paired C1 differences and multiplicity-adjusted tests.

Fig. 2 shows the descriptive window trajectories used to localize, but not test, temporal changes.

B. Registry and Prototype Diagnostics

The direct registry contrast does not validate early cross-silo transfer. For IBM, one- and three-window post-onset recall differences are exactly zero in all ten seeds. Across the full post-onset horizon the registry recovers only one additional target transaction over 1,731 event-positive seed-run records: the mean difference is 0.0006, with one win and nine ties ($p_H = 1.000$). On SAML-D, both variants recover zero target transactions at every horizon over 5,851 such records. The nonzero mean registry admissions— 1.0 ± 0.0 and 1.0 ± 0.0 per IBM run, and 0.7 ± 0.5 and 1.3 ± 0.7 per SAML-D run—therefore do not translate into a reliable capture gain. FedTypo’s aggregate AUPRC advantage over NoReg cannot by itself establish the intended early-transfer pathway.

A post hoc SAML-D re-scoring sweep over $\beta \in \{0, 0.05, 0.10, 0.15, 0.20, 0.30\}$ uses common trained trajectories. At the preselected $\beta = 0.15$, mean AUPRC rises over $\beta = 0$ by 10.6% in C0 and 49.8% in C1. The C1 contrast survives Holm correction across the five nonzero settings ($p_H = 0.039$), but C0 does not ($p_H = 0.312$); C1 peaks at 0.10 and is similar through 0.30. P@50 changes at 0.15 are nonsignificant in C0/C1 ($p_H = 0.938/0.750$). The complete sweep is in the artifact. Thus the term affects drift-stream ranking without establishing earlier cross-silo capture.

Final prototypes show weak correspondence to named IBM typologies. In C0, all-positive purity/NMI/ARI are 0.510/0.121/0.054, but named-only values fall to 0.252/0.054/0.020 and the same-minus-different cosine gap is -0.018. The C1 pattern is similar. SAML-D has moderate agreement: C0 purity/NMI/ARI are 0.404/0.378/0.192

with cosine gap 0.139, declining slightly under C1. These diagnostics support a behavioral-clustering interpretation, especially on IBM, rather than recovery of named laundering schemes. Fig. 3 summarizes prototype fidelity and temporal support.

C. Partition and Budget Sensitivity

Table V reports the outcome-conditioned partition stress test.

Under typology skew, FedTypo AUPRC is 0.0087 and 0.0083 on IBM, and 0.0065 and 0.0079 on SAML-D. These effects are much larger than under account hashing, showing that outcome-conditioned silo construction materially changes the apparent advantage. They are reported only as a stress test.

Fig. 4 varies the alert budget without retraining. The method ordering at P@50 is not the same as for primary AUPRC: Local is best at the prespecified cutoff in both primary-partition datasets. Accordingly, budget curves are operational diagnostics rather than confirmation of the primary ranking result.

D. Component Ablations

The six ablations vary grouping count, random grouping, MMD stability, aggregation weighting, and global coupling. Across two datasets and two conditions, none of the 24 two-sided comparisons with FedTypo-NoReg remains significant after Holm correction; the smallest adjusted value is 0.504. The experiment therefore does not isolate any one of these components as the cause of the NoReg behavior. FedAvg and the window-adapted CDA-FedAvg also produce identical seed-level outputs in every dataset, condition, and reported metric, indicating that the detector/rehearsal control was inactive or behaviorally equivalent under this protocol. Fig. 5 reports all paired component contrasts.

VII. DISCUSSION AND LIMITATIONS

Implications for financial forensics. The primary experiments show that FedTypo can improve stream-level ranking, most consistently on IBM, but the evidence does not identify the proposed registry pathway as the cause. The lack of a reliable post-onset capture gain, nonsignificant component ablations, and weak IBM prototype fidelity require a behavioral-ranking interpretation rather than a claim that named laundering schemes are learned or transferred early. Moreover, the AUPRC gain does not carry to the prespecified P@50 cutoff, where Local is strongest in both primary datasets.

Synthetic evidence. Both benchmarks are synthetic. They provide complete typology ground truth that is rarely available in bank data, but they cannot reproduce investigator policy, hidden undetected laundering, institutional topology, or adaptive offenders. C1 re-times known positives; it evaluates support shift, not zero-day typology generation. Confirmation delay and alert quality are simulations, and a cycle corresponds to different calendar time in the two datasets.

Partition and scale. The primary account-hash partition is label-independent, online-constructible, and keeps each source

TABLE II
IBM AMLWORLD SEED MEANS [95% BOOTSTRAP CI] OVER TEN SEEDS; AUPRC IS PRIMARY AND P@50 SECONDARY.

Method	AUPRC [95% CI]	P@50 [95% CI]
<i>C0: natural control stream</i>		
Local	0.0013 [0.0012, 0.0013]	0.0142 [0.0127, 0.0158]
FedAvg	0.0013 [0.0012, 0.0013]	0.0131 [0.0112, 0.0151]
FedProx	0.0013 [0.0012, 0.0013]	0.0131 [0.0110, 0.0152]
FedProto	0.0012 [0.0012, 0.0013]	0.0125 [0.0106, 0.0145]
CDA-FedAvg	0.0013 [0.0012, 0.0013]	0.0131 [0.0112, 0.0151]
FedTypo-NoReg	0.0013 [0.0012, 0.0013]	0.0129 [0.0110, 0.0150]
FedTypo	0.0036 [0.0033, 0.0040]	0.0137 [0.0124, 0.0151]
<i>C1: abrupt + staggered typology onsets</i>		
Local	0.0013 [0.0012, 0.0013]	0.0143 [0.0127, 0.0160]
FedAvg	0.0013 [0.0012, 0.0013]	0.0130 [0.0112, 0.0149]
FedProx	0.0013 [0.0012, 0.0013]	0.0130 [0.0110, 0.0152]
FedProto	0.0012 [0.0012, 0.0013]	0.0128 [0.0108, 0.0148]
CDA-FedAvg	0.0013 [0.0012, 0.0013]	0.0130 [0.0112, 0.0149]
FedTypo-NoReg	0.0013 [0.0012, 0.0013]	0.0129 [0.0110, 0.0150]
FedTypo	0.0037 [0.0032, 0.0042]	0.0135 [0.0120, 0.0150]

TABLE III
SAML-D RESULTS UNDER THE SAME ALGORITHM AND INFERENCE PROTOCOL. THE 17-DAY WINDOW DURATION RETAINS THE COMPLETE STREAM.

Method	AUPRC [95% CI]	P@50 [95% CI]
<i>C0: natural control stream</i>		
Local	0.0016 [0.0012, 0.0021]	0.0336 [0.0274, 0.0402]
FedAvg	0.0013 [0.0012, 0.0015]	0.0150 [0.0077, 0.0231]
FedProx	0.0014 [0.0012, 0.0017]	0.0120 [0.0058, 0.0198]
FedProto	0.0013 [0.0011, 0.0016]	0.0081 [0.0038, 0.0130]
CDA-FedAvg	0.0013 [0.0012, 0.0015]	0.0150 [0.0077, 0.0231]
FedTypo-NoReg	0.0013 [0.0012, 0.0016]	0.0142 [0.0094, 0.0195]
FedTypo	0.0015 [0.0013, 0.0017]	0.0132 [0.0087, 0.0184]
<i>C1: abrupt + staggered typology onsets</i>		
Local	0.0018 [0.0012, 0.0028]	0.0316 [0.0258, 0.0377]
FedAvg	0.0013 [0.0011, 0.0016]	0.0138 [0.0091, 0.0190]
FedProx	0.0013 [0.0011, 0.0016]	0.0124 [0.0062, 0.0196]
FedProto	0.0017 [0.0012, 0.0023]	0.0101 [0.0055, 0.0149]
CDA-FedAvg	0.0013 [0.0011, 0.0016]	0.0138 [0.0091, 0.0190]
FedTypo-NoReg	0.0013 [0.0011, 0.0015]	0.0137 [0.0079, 0.0200]
FedTypo	0.0020 [0.0015, 0.0027]	0.0162 [0.0100, 0.0225]

TABLE IV
PAIRED C1 AUPRC DIFFERENCES, RANK-BISERIAL EFFECTS, WILCOXON p , AND WITHIN-PANEL HOLM p_H OVER TEN SEEDS.

Baseline	Mean difference [95% CI]	r_{rb}	p	p_H
<i>IBM AMLworld</i>				
Local	0.0024 [0.0019, 0.0029]	1.000	0.002	0.012
FedAvg	0.0024 [0.0020, 0.0029]	1.000	0.002	0.012
FedProx	0.0024 [0.0020, 0.0029]	1.000	0.002	0.012
FedProto	0.0024 [0.0020, 0.0029]	1.000	0.002	0.012
CDA-FedAvg	0.0024 [0.0020, 0.0029]	1.000	0.002	0.012
FedTypo-NoReg	0.0024 [0.0020, 0.0029]	1.000	0.002	0.012
<i>SAML-D</i>				
Local	0.0001 [-0.0002, 0.0004]	0.491	0.193	0.387
FedAvg	0.0007 [0.0003, 0.0012]	0.964	0.004	0.020
FedProx	0.0007 [0.0003, 0.0012]	1.000	0.002	0.012
FedProto	0.0002 [-0.0004, 0.0008]	0.455	0.232	0.387
CDA-FedAvg	0.0007 [0.0003, 0.0012]	0.964	0.004	0.020
FedTypo-NoReg	0.0007 [0.0003, 0.0012]	1.000	0.008	0.023

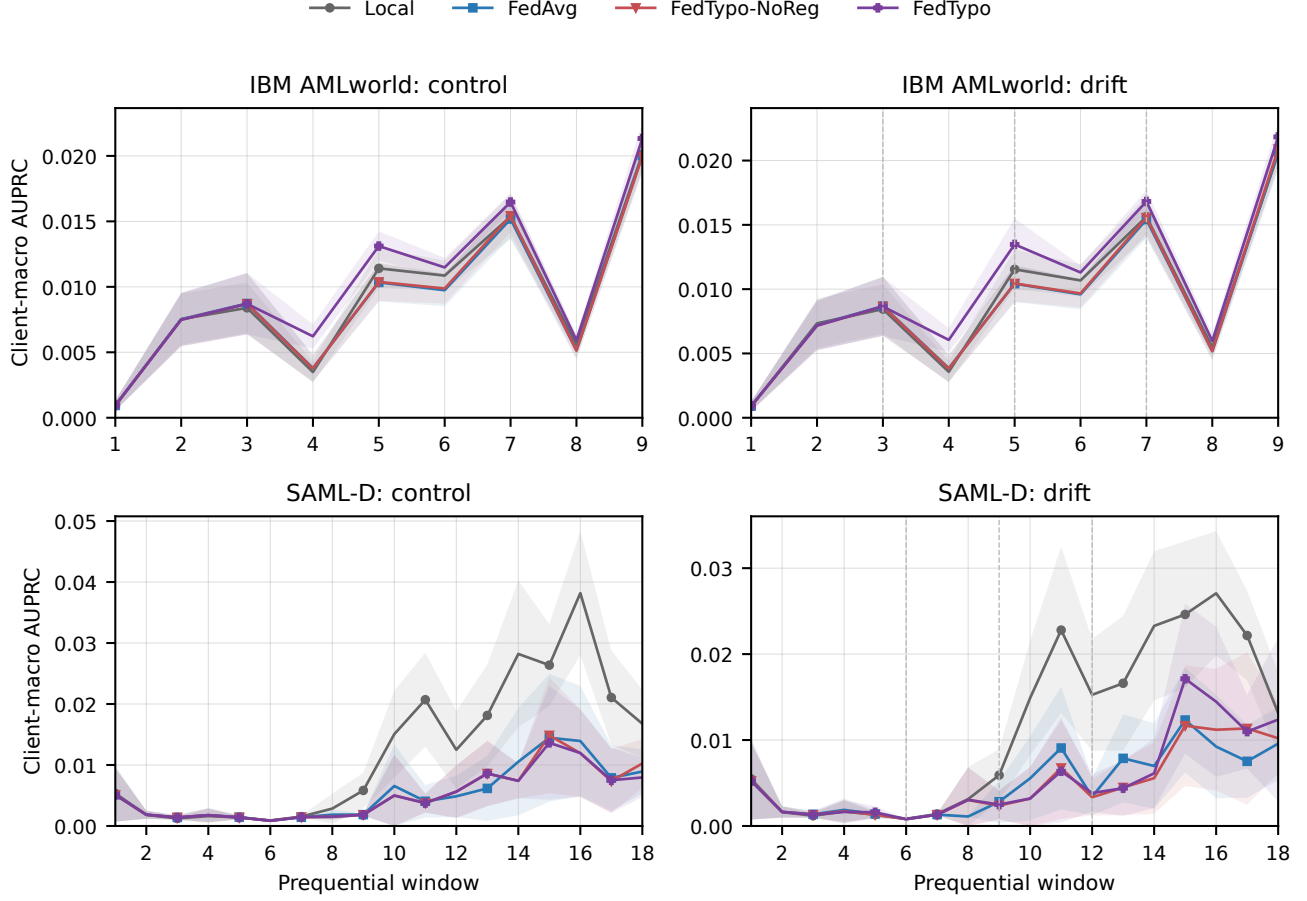


Fig. 2. Positive-only client-window AUPRC trajectories after the warm-up window. These descriptive curves localize temporal variation; they are not the primary stream-level estimand or the inferential unit. Lines show means over ten seeds, bands show descriptive 95% normal intervals, and dashed lines indicate controlled C1 onsets.

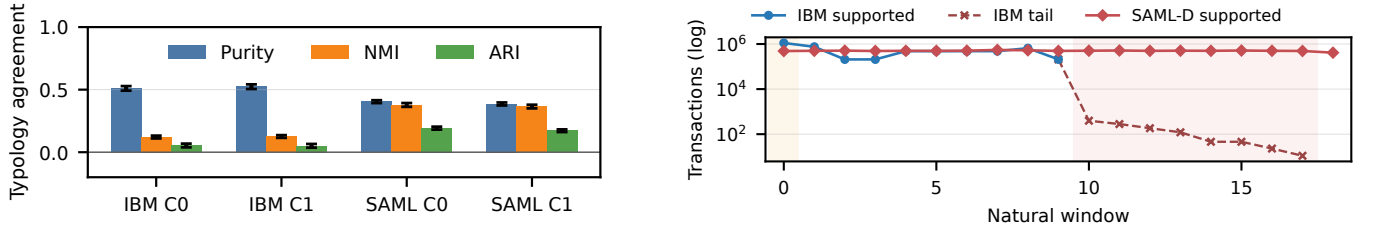


Fig. 3. Mechanism diagnostics. Left: agreement between final prototype assignments and annotated positive typologies. Right: natural-window transaction support motivating the common support rule. Error bars are descriptive intervals over the aggregation units indicated in each panel.

account in one silo, but random hashing is not evidence that five simulated clients match five banks. The typology-skew sensitivity is deliberately transductive and outcome-conditioned; its substantially larger apparent effects demonstrate partition dependence and cannot be promoted to the primary claim. Results may change with natural institutional boundaries, more clients, partial participation, or client arrival/departure. Pairwise prototype matching is quadratic in client count.

Registry calibration. The additive registry score is intentionally simple and can miscalibrate local rankings. Its gate, support threshold, and boost are fixed across the primary

runs and datasets; the post hoc SAML-D re-scoring sweep is descriptive and does not change the selected value. Its AUPRC plateau under drift reduces concern about a single fortunate boost, but the nonsignificant $P@50$ contrasts and null post-onset capture test still leave the mechanism unvalidated. A production system would require a validation period, expiry and provenance for registry entries, analyst confirmation, and calibration under the institution’s actual alert budget.

Cross-dataset boundary. FedTypo improves primary AUPRC against all comparators on IBM, but SAML-D separates the natural and controlled-drift conditions: no C0 comparison is significant, whereas C1 gains are supported

TABLE V
OUTCOME-CONDITIONED TYPOLOGY-SKEW SENSITIVITY. THIS SECONDARY, TRANSDUCTIVE PARTITION IS NOT USED FOR PRIMARY INFERENCE.

Method	AUPRC [95% CI]	P@50 [95% CI]
<i>IBM C0</i>		
FedAvg	0.0014 [0.0013, 0.0015]	0.0899 [0.0806, 0.0992]
FedTypo-NoReg	0.0014 [0.0013, 0.0015]	0.0897 [0.0809, 0.0984]
FedTypo	0.0087 [0.0078, 0.0097]	0.0859 [0.0789, 0.0929]
<i>IBM C1</i>		
FedAvg	0.0014 [0.0013, 0.0014]	0.0909 [0.0819, 0.0999]
FedTypo-NoReg	0.0014 [0.0013, 0.0014]	0.0901 [0.0821, 0.0982]
FedTypo	0.0083 [0.0077, 0.0090]	0.0879 [0.0804, 0.0954]
<i>SAML-D C0</i>		
FedAvg	0.0055 [0.0041, 0.0071]	0.0584 [0.0515, 0.0648]
FedTypo-NoReg	0.0060 [0.0043, 0.0080]	0.0606 [0.0532, 0.0678]
FedTypo	0.0065 [0.0045, 0.0087]	0.0586 [0.0473, 0.0678]
<i>SAML-D C1</i>		
FedAvg	0.0029 [0.0019, 0.0042]	0.0300 [0.0224, 0.0399]
FedTypo-NoReg	0.0034 [0.0021, 0.0055]	0.0353 [0.0260, 0.0472]
FedTypo	0.0079 [0.0055, 0.0104]	0.0442 [0.0325, 0.0579]

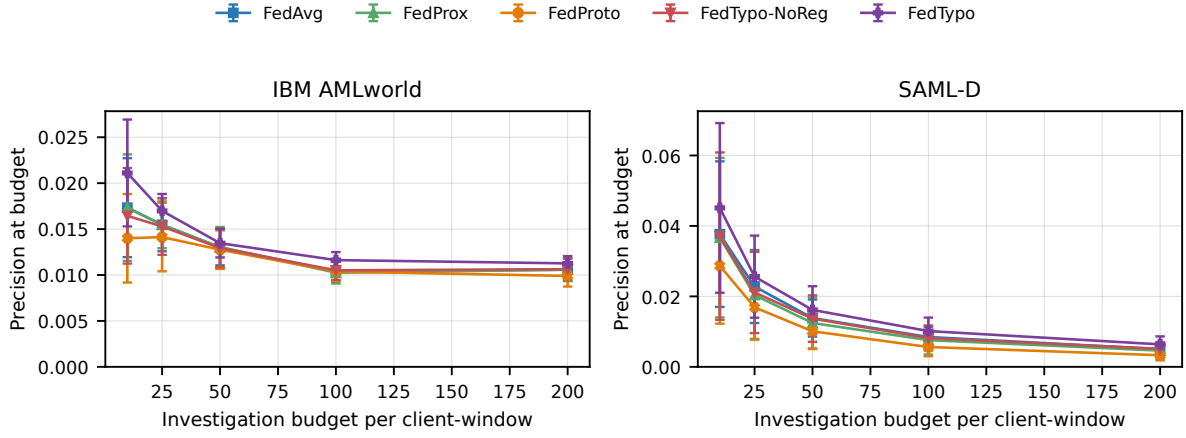


Fig. 4. C1 precision at budgets 10–200, macro-averaged within each seed; error bars are descriptive 95% intervals across seeds.

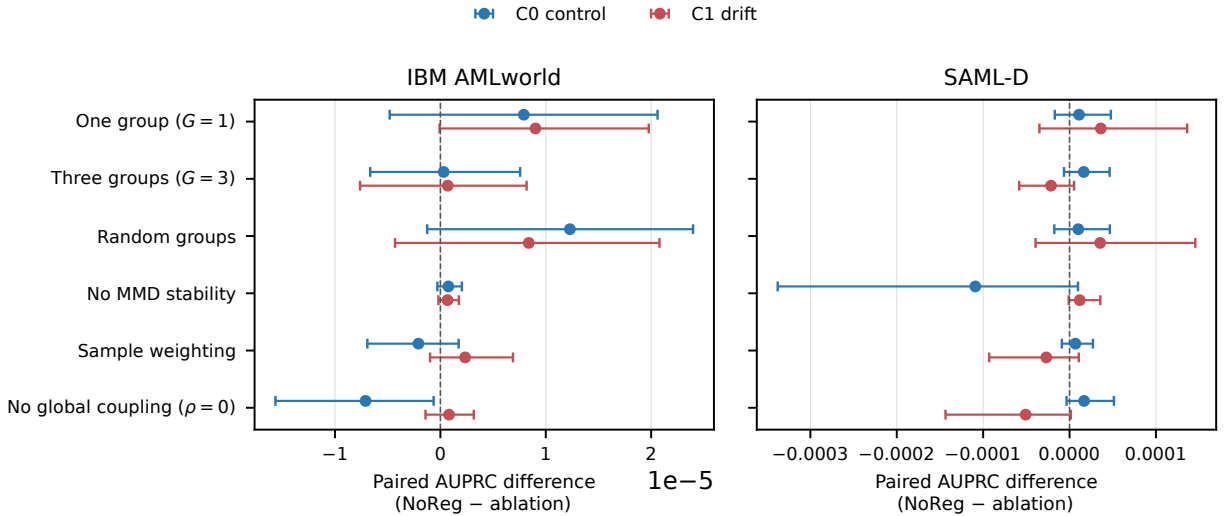


Fig. 5. Component ablations versus FedTypo-NoReg; none of 24 Holm-adjusted paired tests is significant.

against four of six comparators but not Local or FedProto. Neither dataset supports the registry’s intended early-transfer mechanism. The contribution is therefore conditional on the stream, partition, and shift represented by the benchmark, not evidence of a universal prototype-registry advantage.

Security and fairness. A malicious client could poison prototypes or updates, and an honest-but-curious server could attempt inference. Secure aggregation, formal prototype privacy, Byzantine robustness, and auditing of disparate false-positive burden are open requirements. Because the benchmarks do not contain protected attributes or institution-specific investigation outcomes, this study cannot make a fairness claim.

Ethical use. The experiments use public synthetic transactions and no customer records. A deployed score should prioritize human investigation, not automatically deny service or establish guilt; registry entries require provenance, analyst review, appeal, and expiry controls.

VIII. REPRODUCIBILITY AND DATA AVAILABILITY

The artifact contains the canonical experiment script and environment metadata. For every seed it provides client-level, window-level, budget-level, typology-level, post-onset, prototype-level, and paired-difference summaries, plus a deterministic asset builder that generates every numerical macro and figure in this manuscript. The two public datasets are not redistributed: AMLworld HI-Small is available from the IBM AML benchmark release [1], and SAML-D from its authors’ release [20]. The experiment command exposes dataset, partition, output and data roots, seed count, method and condition subsets, run name, optimizer-reset policy, fast-development mode, and optional seed-42 raw-prediction retention through documented environment variables. Library versions and GPU model are written into each consolidated dataset result root. Code, compact result summaries, environment metadata, and deterministic asset generation are publicly available at <https://github.com/PAPPURAJ/FedTypo-AML>. The submission archive contains the same authoritative snapshot. The artifact distinguishes the byte-exact executed source from the current source; an audited postprocessor changes only mutually exclusive win/tie/loss labels, leaving metrics, confidence intervals, effect sizes, and p -values unchanged.

IX. CONCLUSION

FedTypo uses confirmed-positive behavioral prototypes as a common structural interface between federated heterogeneity and delayed temporal adaptation. Prototype-set similarity determines grouped aggregation, immediate MMD tempers unstable updates, and delayed novelty controls a cross-institution registry. A supported-window prequential evaluation on two million-scale AML benchmarks uses ten seeds, label-independent account-hash silos, stream-level client-macro AUPRC, and seed-level inference. FedTypo improves primary AUPRC consistently on IBM and selectively under SAML-D drift, yet Local is strongest at $P@50$. The registry ablation, post-onset capture test, prototype fidelity, and component ablations do not validate the proposed early-transfer

mechanism. Accordingly, FedTypo is a data-local collaborative ranking design whose mechanism and alert-budget value require further validation, not a universally superior detector or a formal privacy or adversarial-security guarantee.

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